

Robot Dynamics & Control Exam

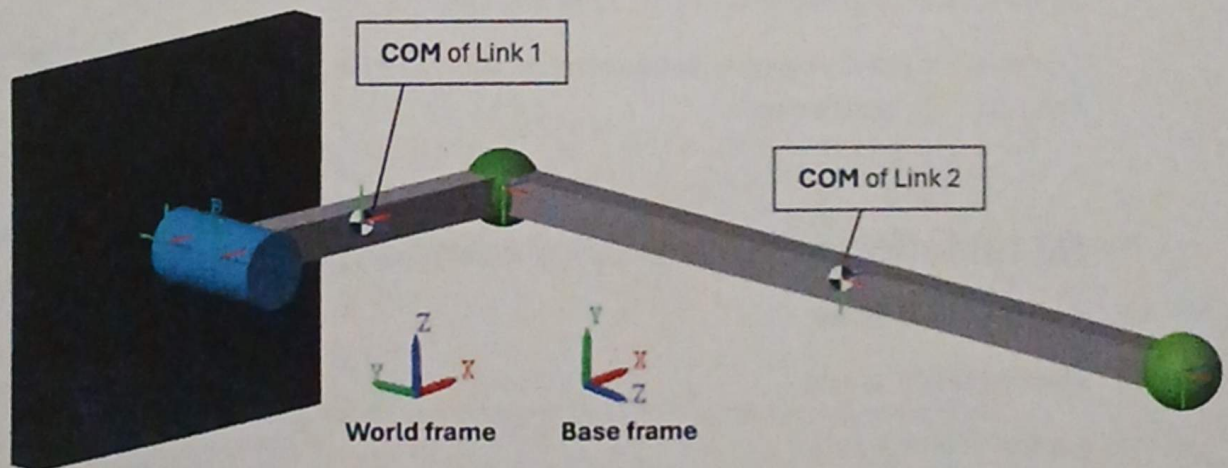
June, 4th, 2025

Duration – 3:00 h

Important notes:

- Regarding the questions about the **Simulink part**, each exercise is marked with an 'S' followed by a number, which indicates the corresponding question number inside the quiz;
- Your Simscape models **must** have the **same configurations** as the figures, with the **same** parameters shown;

2R Non-Planar Robot (2R NP)



For each of the 5 exercises in the Simulink section, you must start from the non-planar 2R robot model shown in the figure, which follows the **Denavit-Hanenberg convention**. The **base frame** is rotated by -90° around the x-axis with respect to the **world frame** (the frame configuration is shown in the figure). As for the frame of **Joint 2** with respect to the **Joint 1**, it is obtained through two rotations: one of $+90^\circ$ around the y-axis, and the second of -180° around the z-axis. For each of the 5 exercises, **keep gravity enabled** in the simulation; disable it only for debugging purposes. For each of the 5 exercises, **the gravity acts along the opposite direction of the z-axis of the world frame**.

S1 question (static gravity compensation)

Parameters of 2R NP

Mass of Link 1: 2.5 kg

Length of Link 1: 0.1 m

Mass of Link 2: 4.0 kg

Length of Link 2: 0.25 m

Mass of sphere 1 and 2: 0kg

Initial Joint configurations: $q_i = [q_1, q_2] = [-45^\circ, 45^\circ]$

Compute the **gravity compensation torque** (expressed in $N \cdot m$) of both joints with respect to the **base frame**.

$$\begin{aligned} \tau_1 &= 1.328 \text{ N.m} \\ \tau_2 &= -2.453 \text{ N.m} \end{aligned}$$

S2 question (static gravity compensation)

Parameters of 2R NP

Mass of Link 1: 2.5 kg

Length of Link 1: 0.1 m

Mass of Link 2: 4.0 kg

Length of Link 2: 0.25 m

Mass of sphere 1: 0kg

Mass of sphere 2 (end effector): 3kg

Initial Joint configurations: $q_i = [q_1, q_2] = [-45^\circ, 45^\circ]$

Compute the **gravity compensation torque** (expressed in $N \cdot m$) of both joints with respect to the **base frame**.

$$\begin{aligned} \tau_1 &= -0.2392 \text{ N.m} \\ \tau_2 &= -6.205 \text{ N.m} \end{aligned}$$

S3 question (Newton-Euler recursive dynamics)

Parameters of 2R NP

Mass of Link 1: 2.5 kg

Length of Link 1: 0.1 m

Mass of Link 2: 4.0 kg

Length of Link 2: 0.25 m

Mass of sphere 1 and 2: 0kg

Initial Joint configurations: $q_i = [q_1, q_2] = [-20^\circ, 50^\circ]$

Width of Link 1 : 0.01 m

Height of Link 1 : 0.01 m

Width of Link 2 : 0.01 m

Height of Link 2 : 0.01 m

Moment of inertia of Link1 and Link 2: You can consider only i_{zz1} and i_{zz2} , since the others are close to 0 kg·m²

The robot must be able to reach the following final configuration in **5 seconds**:

$q_f = [30^\circ, -70^\circ]$. Inside the **polynomial trajectory block** utilize these conditions:

- put a **cubic polynomial trajectory**;
- set the following **$q_dot_desired$** : Joint 1 [0, 0.1] rad/s, Joint 2: [0, 0.2] rad/s.

Do not consider the **damping effects** of both joints.

Compute the **x-component** of the **Inertial force (expressed in N)** of the last link, projected onto its own frame.

S4 question (Newton-Euler recursive dynamics)

Parameters of 2R NP

Mass of Link 1: 2.5 kg

Length of Link 1: 0.1 m

Mass of Link 2: 4.0 kg

Length of Link 2: 0.25 m

Mass of sphere 1 and 2: 0kg

Initial Joint configurations: $q_i = [q_1, q_2] = [-20^\circ, 50^\circ]$

Width of Link 1 : 0.01 m

Height of Link 1 : 0.01 m

Width of Link 2 : 0.01 m

Height of Link 2 : 0.01 m

Moment of inertia of Link1 and Link 2: You can consider only I_{zz1} and I_{zz2} , since the others are close to 0 kg·m²

The robot must be able to reach the following final configuration within **5 seconds**:

$q_f = [30^\circ, -70^\circ]$. Inside **the polynomial trajectory block** utilize these conditions:

- put a **cubic polynomial trajectory**;
- set the following **$q_dot_desired$** : Joint 1 [0, 0.1] rad/s, Joint 2: [0, 0.2] rad/s.

Do not consider the **damping effects** of both joints.

Compute the τ related to **inertial, Coriolis and gravitational parts** (expressed in N · m), of both joints, projected onto the frame of each joint.

$$\begin{aligned}\tau_1 &= 2.721 \\ \tau_2 &= -1.897\end{aligned}$$

S5 question (Computed torque control)

Parameters of 2R NP

Mass of Link 1: 2.5 kg

Length of Link 1: 0.1 m

Mass of Link 2: 4.0 kg

Length of Link 2: 0.25 m

Mass of sphere 1 and 2: 0kg

Initial Joint configurations : $q_i = [q_1, q_2] = [10^\circ, 30^\circ]$

Width of Link 1 : 0.01 m

Height of Link 1 : 0.01 m

Width of Link 2 : 0.01 m

Height of Link 2 : 0.01 m

Moment of inertia of Link1 and Link 2: You can consider only i_{zz1} and i_{zz2} , since the others are close to 0 $\text{kg}\cdot\text{m}^2$

PD controller gains: $K_p=5$; $K_d=2\sqrt{K_p}=4.7$

The robot must be able to reach the following **final configuration** within 10 seconds: $q_f = [90^\circ, 110^\circ]$, considering that at time $t=5\text{s}$ it passes through the **following intermediate configuration**: $q_{int} = [60^\circ, 80^\circ]$.

Inside the **polynomial trajectory block** utilize the following conditions:

- put a cubic polynomial trajectory;
- set the following **$q_{\dot{\text{desired}}}$** : Joint 1 $[0, 0, 0]$ rad/s, Joint 2: $[0, 0, 0]$ rad/s.

Do not consider the **damping effects** of both joints.

Compute the **desired torque** (expressed in $\text{N}\cdot\text{m}$) required to follow the trajectories for both joints, projected onto the frame of each joint.